

Sean Luka Girgis

Python Backend / Data Engineer | APIs, AI/ML Pipelines, AWS & Production Systems

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Professional Summary

Python backend/data engineer with strong enterprise experience building and supporting Python, SQL, PySpark, AWS, API-style integrations, data pipelines, production reporting systems, and AI/ML-adjacent workflows. Experienced turning prototypes and data workflows into reliable production-ready systems with validation checks, operational support, documentation, and stakeholder collaboration. Best aligned to backend roles supporting AI/ML applications, APIs, microservices concepts, data-intensive systems, model-serving readiness, AWS cloud workflows, Docker, and production reliability; FastAPI/Django, Redis, Kubernetes, Azure, and formal MLOps are positioned carefully as adjacent or active-learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built and supported Python/Pandas/PySpark and SQL workflows ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for analytics, reporting, forecasting, validation, and operational decision support.
- Developed Python-based backend/data workflows that integrated multiple internal systems, monitoring feeds, reporting layers, and warehouse-backed services into reliable business-facing outputs.
- Prepared forecasting-ready datasets for Prophet and scikit-learn workflows, supporting capacity planning, trend analysis, bottleneck prediction, and AI/ML-adjacent infrastructure planning.
- Used AWS S3, Glue, Athena, Redshift, EC2/ECS concepts, PySpark, Oracle, and SQL to support scalable processing, cloud-oriented data workflows, and production reporting reliability.
- Supported production systems by tracing failed outputs, source-feed behavior, Python processing, SQL transformations, API-style data integration issues, latency symptoms, and downstream reporting discrepancies.
- Created validation and regression-style checks for row counts, nulls, duplicates, freshness, abnormal deltas, unmatched records, and output comparisons against trusted prior runs.
- Partnered with application, infrastructure, analytics, engineering, and business stakeholders to clarify requirements, prioritize fixes, explain system behavior, and deliver reliable improvements.
- Created runbooks, data flow notes, API/data integration notes, metric definitions, troubleshooting guides, and support procedures to improve maintainability and team handoff.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Supported production-facing Brand.com and hospitality platforms using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.
- Analyzed application behavior, transaction patterns, monitoring signals, and production-like symptoms to identify reliability and performance issues.
- Created dashboards and reporting views that converted raw monitoring and transaction data into actionable performance, reliability, and stakeholder insights.

- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical platform improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services application monitoring environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Provided production troubleshooting guidance to application, middleware, infrastructure, and operations teams using CA APM/Introscope telemetry, dashboards, alerts, and transaction visibility.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable operational processes.
- Created technical documentation, support procedures, onboarding material, dashboard standards, alert thresholds, and troubleshooting notes used by multiple teams.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during performance and production-readiness testing.
- Documented test results, expected behavior, defects, risks, performance metrics, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led validation and production-readiness support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported phased cutover, troubleshooting, documentation, and defect resolution for a mission-critical enterprise transaction platform.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, VC++ components, and DB2-backed integration modules in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using C, C++, Pro*C, Oracle, PL/SQL, SQL scripts, XML, Java/J2EE concepts, and enterprise messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.

- Provided production support, defect investigation, data maintenance, issue escalation, troubleshooting, and release support for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with FastAPI service layers, Pydantic request/response contracts, Docker-first smoke tests, deterministic API endpoints, and testable service behavior.

Technologies: Python, FastAPI, Docker, Pydantic, pytest, REST API Concepts

HorizonScale — Capacity Forecasting and Data Platform

Built Python/Pandas/PySpark workflows to process telemetry, validate outputs, prepare forecasting datasets, and support AI/ML-driven capacity planning and bottleneck prediction.

Technologies: Python, Pandas, PySpark, SQL, Prophet, scikit-learn

Serverless Lakehouse Platform (AWS)

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet with validation checkpoints, SQL quality checks, and curated analytics datasets.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL

Production Data Quality and RCA Workflow

Built validation and troubleshooting workflows for production reporting datasets, including row-count checks, missing feed diagnosis, unmatched join checks, prior-run comparisons, and documented recovery steps.

Technologies: Python, SQL, Pandas, Oracle, Data Validation, RCA